Transaction Processing Manual

BASE24-atm®

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Contents

Introduction *continued*

BASE24-atm Transaction Processing Controls *continued*

ACI Worldwide, Inc.

Preface

The ***BASE24-atm Transaction Processing Manual*** describes how BASE24-atm processes transactions. It includes basic transaction processing concepts and definitions, as well as specifics on how transactions are identified and processed by BASE24-atm. It also describes how to control and configure various aspects of BASE24-atm transaction processing.

Audience

This manual is intended for the BASE24 managers and operational staff who are responsible for configuring the BASE24-atm database for transaction processing and understanding how the transactions flow through the BASE24-atm system.

Prerequisites

Persons using this manual should be familiar with the BASE24-atm product, its basic functions, its terminology, and the BASE24 files it uses.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information.

This manual contains references to the following BASE24 publications:

- The ***BASE24 Logical Network Configuration File Manual*** describes, in detail, each assign and param used in configuring BASE24-atm Device Handler, Authorization, and Host Interface processes.
- The ***BASE24 Base Files Maintenance Manual*** and ***BASE24-atm Files Maintenance Manual*** provide detailed descriptions and valid values for the configuration fields discussed in this manual.
- The ***BASE24 Files and Record Structures Manual*** provides general information about the structure of BASE24 files and records, along with instructions for using the ACI DDLDOC program.
- The ***BASE24 Tokens Manual*** presents detailed field descriptions and configuration information for the tokens used by BASE24-atm.
- The ***BASE24 ISO Host Interface Manual, BASE24 ANSI Host Interface Manual, BASE24 BIC ISO Interface Manual,*** and ***BASE24 BIC ANSI Interface Manual*** provide detailed processing information for each of the specified interfaces.
- The ***BASE24 BIC ISO Standards Manual*** contains information about the standards used with the BIC ISO Interface.
- The ***BASE24 External Message Manual*** presents detailed field descriptions and configuration information for the ISO-based external message used between BASE24 and a host. (The ANSI-based external message is documented in the ***BASE24 ANSI Host Interface Manual***.)
- The ***BASE24-atm Settlement and Reporting Manual*** describes, in detail, the settlement and reporting functions supported by BASE24-atm and how to configure them.
- The ***BASE24 Transaction Security Manual*** describes BASE24 support of PIN verification and encryption, card verification, message authentication, and dynamic key management, along with the database settings required to implement this support.
- The ***BASE24-atm NCR 5XXX Device Support Manual*** describes BASE24-atm support of NCR 5XXX-series ATMs.
- The ***NET24-XPNET Network Control Command Reference Manual*** provides detailed information on configuring a BASE24 network of lines, stations, and processes.
- The ***BASE24-atm Diebold 10XX/478X Device Support Manual*** describes BASE24-atm support of Diebold 10XX/478X ATMs.

- The ***BASE24-atm Multiple Currency Support Manual*** describes the BASE24-atm Multiple Currency add-on product.
- The ***BASE24-atm EMV Manual*** describes the BASE24-atm EMV add-on product.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

"Conventions Used in this Manual" follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, "Introduction," introduces the reader to the basics of BASE24-atm transaction processing. It includes definitions and discussions of transaction originators and authorizers, issuers and acquirers, hosts and DPCs, cards and accounts, supported transactions, transaction messages, and transaction logging.

Section 2, "BASE24-atm Transaction Path," discusses the BASE24-atm processes involved in transaction processing and how they are linked together to form various transaction paths. It also describes the timers they use to control processing.

Section 3, "BASE24-atm Transaction Processing Controls," describes the various transaction processing control options that enable a logical network to be tailored to the needs of its institutions. It includes discussions of such topics as authorization levels, authorization methods, sharing, and cardholder limits and accumulation.

Section 4, "BASE24-atm Routing and Authorization," describes—and provides examples of—how BASE24-atm routes transactions and selects authorization parameters for processing transactions.

Section 5, "BASE24-atm Message Flows," presents a series of diagrams illustrating the paths taken by BASE24-atm transactions under various types of processing situations.

Section 6, "BASE24-atm Transaction Impacts on the Database," describes the impacts of each type of BASE24-atm transaction on the Cardholder Authorization File (CAF), Positive Balance File (PBF), BASE24-atm Terminal Data files, and Usage Accumulation File (UAF).

Appendix A, "BASE24-atm Internal Message," describes the fields in the BASE24-atm Standard Internal Message (STM) and identifies the BASE24-atm tokens that can be carried in the internal message.

Appendix B, "BASE24-atm Card Disposition and Authorization Response Codes," describes the response codes that can be carried in the STM to identify the results of processing a transaction.

Appendix C, "BASE24-atm Device Handler Processing," summarizes the transaction processing performed by a BASE24-atm Device Handler process.

Appendix D, "Pregen Utility," describes the Pregen utility and the procedures used to build and start using a new or revised SPROUTE File.

Appendix E, "Base Extended Memory Table Build Utility," provides detailed instructions for using the Base Extended Memory Table Build utility. This utility is used to build the extended memory tables for the Acquirer Processing Code File (APCF), Institution Definition File (IDF), Issuer Processing Code File (IPCF), Receipt File (RCPT), and Terminal Receipt File (TRF).

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, AT-AE000-07 for the ***BASE24-atm Transaction Processing Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Nov-2007 for November, 2007) indicates the issue of the manual. The software release information (for example, R6.0v8 for release 6.0, version 8) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

Conventions Used in This Manual

This section explains how syntax and required blank characters are documented in this manual. It also explains BASE24-atm Terminal Data files acronyms.

Syntax Notation

Syntax descriptions in this manual use several symbols and conventions to denote how commands must be entered. These conventions are described in the table below.

Notation	Meaning
UPPERCASE LETTERS	Indicate key words and literals that you must enter exactly as shown.
lowercase italics	Identify variables that you must provide.
Brackets [ ]	Enclose optional syntax items that you can enter in the command.
Braces { }	Enclose a group of items from which you are required to choose one item. The items are separated within the braces by a vertical bar (|).
Vertical bar |	Separates alternatives in a list of items.
Ellipsis …	Indicates the immediately preceding syntactical item can occur one or more times.
Punctuation	Must be entered as shown. This includes parentheses, commas, semicolons, and other symbols not previously described.

Notation	Meaning
Spaces	Are required as delimiters wherever it would be otherwise impossible to discriminate between adjacent words or symbols. You can insert additional spaces between any adjacent words or symbols, but not within a word or multiple character symbol.
Indented lines	Indicates a continuation of the preceding line of syntax. If the syntax of a command is too long to fit on a single line, each continuation line is indented.

Required Blank Spaces

Throughout this manual when discussing the impact of required blanks or spaces in entered field data, the symbol ƀ is used to denote a required blank character or space.

BASE24-atm Terminal Data Files Acronyms

The BASE24-atm Terminal Data files can be abbreviated using a number of acronyms, depending on the Device Handler process used in the BASE24-atm system, whether you are accessing the file through the BASE24 user access system, or whether the data is static or dynamic. The table below summarizes BASE24-atm Terminal Data files acronyms by Device Handler process:

Device Handler Process	BASE24-atm Terminal Data Files Acronym			
	Screen	**File**	**Static**	**Dynamic**
Diebold 10XX/478X	ATD	ATD	ATDS1	ATDD*x*
NCR 5XXX	ATD	ATD	ATDS1	ATDD*x*
Other	ATD	TDF	Not applicable	Not applicable

In the above table, the acronym ATDD*x*, where *x* is the value 1 or 2, can describe the dynamic data stored in the BASE24-atm Terminal Data files. The data in ATDD1 is general dynamic data. The data in the ATDD2 is token data for the last transaction processed by BASE24 and acquired at this terminal.

To eliminate confusion when referring to the file, the name BASE24-atm Terminal Data files is used instead of a file acronym.

ACI Worldwide, Inc.

1: Introduction

This section provides an introduction to BASE24-atm transaction processing. It provides readers with some basic information on how BASE24-atm processes transactions.

BASE24-atm Transaction Processing

BASE24-atm provides the means by which a transaction originator can request and receive authorization for an action on a customer card or account from a transaction authorizer. The role of BASE24-atm in transaction processing includes the following:

- Routing transactions from the transaction originator to the appropriate authorizer.
- Participating in or carrying out authorization on behalf of the institutions for which it processes.
- Logging all transactions for later processing and settlement.

BASE24-atm can accept transactions from a variety of sources. Likewise, it can involve a number of different authorizers in its processing. The following describes the various transaction originators and authorizers that can exist in a BASE24-atm system.

Transaction Originators

BASE24-atm can accept transactions for authorization from several different types of originators. These originators are described below.

ATMs

Automated teller machines (ATMs) can be directly attached to BASE24-atm, in which case transactions are originated by customers or service personnel at the ATM. Communications between the ATM and BASE24-atm are controlled by BASE24-atm Device Handler processes.

Hosts

Host computers can be set up to control ATM networks and forward transactions to BASE24-atm. In this case, communications between these acquirer hosts and BASE24-atm are controlled by BASE24-atm Host Interface processes.

Interchanges

Interchanges can act as switches to forward transactions to BASE24-atm for authorization. In this case, communications between the interchange acquiring the transaction and BASE24-atm are controlled by BASE24-atm Interchange Interface processes.

Co-Networks

Co-networks can be linked to a BASE24-atm system through a BASE24 Interchange (BIC) Interface process. In this case, transactions can be originated in any of the ways noted above and forwarded through BASE24 into another BASE24 system as if through an interchange. Communications between the BASE24 networks is controlled by BASE24 BIC Interface processes on each BASE24 system.

Transaction Authorizers

BASE24-atm can authorize transactions or route them to several types of authorizers. The possible transaction authorizers are described below.

BASE24-atm

BASE24-atm can be set up to authorize transactions in those situations where cardholder files are maintained on the BASE24-atm system.

Hosts

Hosts can be set up to authorize transactions in situations where cardholder files are maintained on the host computer. In this case, BASE24-atm can screen the transactions before sending them to the host and can also be set up to stand in and authorize transactions for a host in situations where communications between BASE24-atm and the host system are down.

Interchanges

Interchanges can be used to forward transactions to other networks for authorization.

Co-Networks

BASE24-atm can be linked to co-networks through a BIC Interface process, and transactions can be forwarded through BASE24 to another BASE24 system or co-network for authorization. BASE24-atm can also be set up to stand in and authorize transactions for another co-network in situations where communications between BASE24-atm and the co-network are down.

Customer-Specific Application Processes

The BASE24-atm Diebold 10XX/478X and NCR 5XXX Device Handler processes and certain Interchange Interface processes can route transactions directly to a process running outside of the BASE24-atm environment (using a customer-specific XPNET process) for authorization. For example, an ATM bill payment transaction acquired by an enhanced endpoint could be sent directly to a bill payment service. The authorization response returned from this application process can then be optionally logged to the Transaction Log File (TLF).

The following Interchange Interface processes support this functionality:

- Banknet Interchange Interface process
- BIC ISO Interface process
- MDS/MDSM Interchange Interface process
- PLUS ISO Interchange Interface process
- VisaNet Interchange Interface process

Issuers and Acquirers

Transaction originators and authorizers are generally defined by the EFT industry terms acquirer and issuer. An acquirer, or transaction acquirer, is the party that originates a transaction. An issuer, or card issuer, is an institution that issues cards or owns accounts that can be used in EFT transactions.

Whether an entity is defined as an issuer or an acquirer depends on the perspective from which the transaction is viewed. Intermediary networks and processors that receive transactions from a transaction originator and authorize them or forward them elsewhere for authorization can be acting on behalf of both the acquirer and the issuer.

From the acquirer point of view, intermediary networks are acting on behalf of the issuer. For example, a cardholder initiates a transaction at an ATM belonging to institution A. In this case, institution A is the transaction acquirer. Institution A sends the transaction to an intermediary network that, in turn, sends the transaction to institution B for authorization. Institution B owns the cardholder's account and issued the cardholder the card; therefore, institution B is the true issuer. However, from the point of view of institution A, the intermediary network is performing as an issuer on behalf of institution B since institution A had to send the transaction to the intermediary network before it could reach its authorizing destination.

From the issuer point of view, intermediary networks are acting on behalf of the acquirer. As in the above example, institution B views the intermediary network as an acquirer since the intermediary network is the entity that sent it a transaction to be authorized.

BASE24 can process transactions on behalf of an issuer or an acquirer, depending on where a transaction originates and who is to authorize the transaction. For example, when BASE24 sends a transaction to a back-end host for authorization, BASE24 represents the acquirer in the message exchange and the back-end host represents the issuer. On the other hand, when a transaction is sent to BASE24 for authorization, the sending host represents the acquirer and BASE24 represents the issuer.

Hosts and DPCs

Hosts can play a prominent role in the transaction processing performed by BASE24. A host is an external mainframe computer system with which BASE24-atm interacts—either online or by batched tape—to authorize transactions and/or update cardholder records.

Although the term host generally refers to an actual computer or system of computers, the term also refers to the institution or organization responsible for the host computer system and its processing. For example, in the phrase, “the host can opt to receive advice messages,” the term host actually refers to the BASE24-defined institution in control of the host computer, rather than the computer itself.

BASE24 also refers to hosts as data processing centers (DPCs). Generally, the terms DPC and host are synonymous. However, the term host implies either a host computer or the organization responsible for a host computer; the term DPC always characterizes the host computer or computer network itself.

A DPC can be one mainframe computer communicating with BASE24, or a DPC can represent several mainframe computers networked together at a host site. Regardless of the machine configuration in relation to BASE24, a DPC represents a single host from the BASE24 point-of-view. A host site can establish multiple DPCs; however, BASE24 considers each of these DPCs to be a separate host.

DPCs are assigned identification numbers and are defined to BASE24 in the Host Configuration File (HCF). Each DPC must have a record in the HCF to be recognized by BASE24. The HCF contains options at the DPC-level that allow hosts to specify individual processing parameters for each DPC.

Cards and Accounts

Transactions in the BASE24-atm system can be defined as specific actions—usually financial—carried out on accounts of some type using a card or a card number as a basis for initiating the action.

BASE24 makes a distinction between cards and accounts. Transaction limits and accumulators can be maintained at the card or account level. Balances, if maintained, are maintained at the account level. The following discusses cards and accounts as they are defined and used in BASE24.

Cards

Plastic cards serve as evidence of an account and as a mechanism for accessing the account from electronic funds transfer (EFT) devices such as automated teller machines (ATMs) and point-of-sale (POS) terminals. Cards are given card numbers, which may or may not match the account numbers of the accounts that they are used to access. There can be a one-to-one relationship between cards and accounts, one card can access multiple accounts, or one account can be accessed by multiple cards, depending on card issuance procedures and BASE24 parameters.

Credit Cards and Debit Cards

In BASE24-atm, credit cards and debit cards can access both noncredit and credit accounts. BASE24-atm does not distinguish between credit and debit cards for the purpose of transaction processing.

Card Type Codes

BASE24 uses codes to identify the type of card being used to initiate transactions. Card type codes are either reserved by BASE24 or user-defined.

Reserved BASE24 Card Types

BASE24 reserves specific card types across all product lines. These reserved codes are to be used only as defined. Refer to the ***BASE24 Base Files Maintenance Manual*** for a list of reserved card types.

User-Defined Card Types

Users can define and use any one- or two-character code, not reserved by BASE24, according to the following guidelines:

- The first character must be alphabetic: A, B, D through O, and Q through Z.
- The second character can be A through Z, 0 through 9, or a blank.

A valid COBNAMES table entry is recommended for each user-defined code.

Note: Codes starting with C and P can also be user-defined, but these are subject to restrictions imposed by BASE24 (refer to the reserved BASE24 card types list above). The BASE24 standard COBNAMES table already contains entries for these codes; however, the COBNAMES entries can be modified as needed by the user. Refer to the ***BASE24 Base Files Maintenance Manual*** for more information about card types.

Accounts

BASE24 classifies all accounts as one of two types: credit or noncredit. This classification determines the account-level limits and accumulators that apply to a particular transaction.

Credit and Noncredit Accounts

Credit accounts involve funds advanced to an accountholder, by a financial institution or retailer, based on a credit agreement with the accountholder.

Noncredit accounts involve accountholder funds on deposit with a financial institution (for example, checking or savings).

BASE24 allows processors to set up various transaction limits and accumulators. Throughout BASE24, the account classification determines which limits and accumulators apply to a particular transaction.

Account Type Codes

BASE24-atm identifies accounts by account type as described in the table below. Institutions can set up other account types in the account type ranges, but BASE24-atm treats these account types as if they are the first in the number range. For example, an institution may want to distinguish between different types of checking accounts for their own record keeping and reporting purposes. They can assign type 02 to checking account type A, 03 to checking account type B, and 04 to checking account type C. However, during processing, BASE24-atm treats account types 01, 02, 03, and 04 the same.

Type	Account Description	Credit	Noncredit
01	Checking (DDA)		✓
02–09	Checking (DDA) as set up by the institution. These types are processed the same as type 01.		✓
11	Savings		✓
12–19	Savings as set up by the institution. These types are processed the same as type 11.		✓
31	Credit	✓	
32–39	Credit account as set up by the institution. These types are processed the same as type 31.	✓	
60	Other. Accounts of this type are used with multiple account selection by account qualifier.	✓	✓

Account Type Names

Screens for certain BASE24 files use account type names defined in the Account Type Table File (ATT) instead of the account type codes listed above. Account type names are used on Acquirer Processing Code File (APCF), Issuer Processing Code File (IPCF), and Terminal Receipt File (TRF) screens, which BASE24-atm uses for defining a set of transactions allowed and the accounts on which the transactions can be performed based on the processing code and transaction profile.

ACI provides a default ATT that contains account type names for a number of account type codes. The ATT uses account type codes based on the International Organization for Standardization (ISO), rather than the account type codes used internally by BASE24-atm. The following table shows how ISO and BASE24 account type codes relate to account type names provided in the default ATT. The first two columns of the table list the ISO and BASE24-atm account type codes, respectively. The third column lists the corresponding account type name in the ATT, and the last column describes the account.

Account Type		Account Type Name	Description
ISO	BASE24		
00	Not applicable	NONE	
10	11	SAV	Savings accounts
1A	14	SAV1	
1B	15	SAV2	
1C	16	SAV3	
1D	17	SAV4	
1E	18	SAV5	
1F	19	SAV6	
20	01	DDA	Demand deposit (checking) accounts
2A	02	DDA1	
2B	03	DDA2	
2C	04	DDA3	
2D	05	DDA4	
2E	06	DDA5	
2F	07	DDA6	
2G	08	DDA7	
2H	09	DDA8	

Account Type		Account Type Name	Description
ISO	BASE24		
30	31	CR	Credit accounts
3A	33	CR1	
3B	34	CR2	
3C	35	CR3	
3D	36	CR4	
3E	37	CR5	
3F	38	CR6	
3G	39	CR7	
9M	60	OTHER	Other accounts used with multiple account selection by account qualifier processing

BASE24-atm Transactions

BASE24-atm supports the transactions described on the following pages. Unless noted to the contrary, BASE24-atm supports these transactions in all respects. However, not all ATMs and interchanges support all of the listed transactions. Refer to the appropriate BASE24-atm device support manual for a complete list of transactions supported by a particular device. Refer to the appropriate Interchange Interface documentation for a complete list of transactions supported by a specific Interchange Interface process.

Transaction Identifiers

Transactions are identified in BASE24-atm by a series of three values known as the processing code. The processing code is carried in the TRAN-CDE, FROM-ACCT-TYP, and TO-ACCT-TYP fields of the BASE24-atm Standard Internal Message (STM), and consists of a transaction code, *from* account type, and *to* account type. For more information about these STM fields, refer to appendix A.

Note: Instead of a *from* account and a *to* account, the transaction code for a split deposit transaction contains a first *to* account, and the second *to* account. This transaction is only available through NCR and Diebold self-service banking (SSB).

The following pages provide the six-digit transaction code and a brief description for each of the BASE24-atm transactions, with similar transactions grouped together.

In addition, some transactions have subtypes associated with them. Transaction subtypes enable BASE24-atm customers to distinguish among the types of transactions that have identical processing codes.

Withdrawal Transactions

BASE24-atm supports the following withdrawal transactions.

Note: Subtype APT0 is used to identify a preferred transaction.

Tran Code	Transaction Type and Description
10 00 00	**Fast Cash Withdrawal —** The withdrawal of a predetermined amount of money or purchase of a noncurrency item from a predefined customer account. This transaction reduces the cardholder's selection process to one or two simple key entries at the ATM, decreasing the overall transaction time. **Note**: The *from* account type for fast cash withdrawals cannot be set to 00 if the IDF routing parameters differ by account type.
10 01 00	**Withdrawal from Checking Account —** A cash withdrawal from a cardholder's checking account. This transaction type includes the purchase of a noncurrency item from the ATM.
10 11 00	**Withdrawal from Savings Account —** A cash withdrawal from a cardholder's savings account. This transaction type includes the purchase of a noncurrency item from the ATM.
10 31 00	**Withdrawal from Credit Account —** The advance of funds from a cardholder's credit account. This transaction type includes the purchase of a noncurrency item from the ATM.
10 60 00	**Withdrawal from Other Account —** A cash withdrawal or advance of funds from an account identified by an account qualifier.

Deposit Transactions

BASE24-atm supports the following deposit transactions.

Notes: BASE24-atm self service banking (SSB) allows a check to be deposited either in an envelope or directly into the ATM. The transaction code is the same for either method.

In addition, subtype APT0 is used to identify a preferred transaction.

Tran Code	Transaction Type and Description
20 00 01	**Deposit to Checking —** An unverified deposit of funds to a cardholder's checking account.
20 00 11	**Deposit to Savings —** An unverified deposit of funds to a cardholder's savings account.
20 00 60	**Deposit to Other —** An unverified deposit of funds to an account identified by an account qualifier.
24 00 01	**Deposit to Checking with Cash Back —** An unverified deposit of funds to a cardholder's checking account, with part or all of the deposit amount back in cash.
24 00 11	**Deposit to Savings with Cash Back —** An unverified deposit of funds to a cardholder's savings account, with part or all of the deposit amount back in cash.
24 00 60	**Deposit to Other with Cash Back —** An unverified deposit of funds to a cardholder's account identified by an account qualifier, with part or all of the deposit amount back in cash.
20 01 01	**Split Deposit—Two Checking Accounts —** An unverified deposit of funds into two separate checking accounts. This transaction is available only with BASE24-atm self-service banking (SSB).
20 01 60	**Split Deposit—Checking and Other —** An unverified deposit of funds into a checking account and an account identified by an account qualifier. This transaction is available only with BASE24-atm self-service banking (SSB).

Tran Code	Transaction Type and Description
20 01 11	**Split Deposit—Checking and Savings —** An unverified deposit of funds into a checking account and a savings account. This transaction is available only with BASE24-atm self-service banking (SSB).
20 11 01	**Split Deposit—Savings and Checking —** An unverified deposit of funds into a checking account and a savings account. This transaction is available only with BASE24-atm self-service banking (SSB).
20 11 11	**Split Deposit—Two Savings Accounts —** An unverified deposit of funds into two separate savings accounts. This transaction is available only with BASE24-atm self-service banking (SSB).
20 11 60	**Split Deposit—Savings and Other —** An unverified deposit of funds into a savings account and an account identified by an account qualifier. This transaction is available only with BASE24-atm self-service banking (SSB).
20 60 01	**Split Deposit—Other and Checking —** An unverified deposit of funds into an account identified by an account qualifier and a checking account. This transaction is available only with BASE24-atm self-service banking (SSB).
20 60 11	**Split Deposit—Other and Savings —** An unverified deposit of funds into an account identified by an account qualifier and a savings account. This transaction is available only with BASE24-atm self-service banking (SSB).
20 60 60	**Split Deposit—Two Other Accounts —** An unverified deposit of funds into two different accounts identified by account qualifiers. This transaction is available only with BASE24-atm self-service banking (SSB).

Mobile Top-Up Transactions

The following mobile top-up transactions are defined for the definitions-based model. The BASE24-atm Transaction Context Manager (TCM) will log this transaction code representing the secondary service authorization.

Tran Code	Transaction Type and Description
25 00 00	**Mobile Top-Up —** A mobile top-up transaction requests a secondary provider (e.g., telephone company) to load additional funds or minutes to a mobile telephone account. **Note:** The Authorization process handles the funds authorization transaction as a withdrawal (transaction code 10).
26 00 00	**Mobile Top-Up with Funds —** A mobile top-up using funds transaction request is logged to the Transaction Log File (TLF) by the Transaction Context Manager when the TCM is unable to process transactions. **Note:** The Authorization process handles the funds authorization transaction as a withdrawal (transaction code 10).

Balance Inquiry Transactions

BASE24-atm supports the following balance inquiry transactions.

Note: Subtype APT0 is used to identify a preferred transaction.

Tran Code	Transaction Type and Description
30 01 00	**Balance Inquiry from Checking —** An inquiry on the amount in a cardholder’s checking account.
30 01 11	**Balance Inquiry from Checking and/or Savings —** An inquiry on the amount in a cardholder’s checking account and/or savings account.
30 11 00	**Balance Inquiry from Savings —** An inquiry on the amount in a cardholder’s savings account.

Tran Code	Transaction Type and Description
30 11 01	**Balance Inquiry from Savings and Checking —** An inquiry on the amount in a cardholder's savings account and checking account.
30 31 00	**Balance Inquiry from Credit Account —** An inquiry on the amount in a cardholder's credit account.
30 60 00	**Balance Inquiry from Other —** An inquiry on the amount in a cardholder's account that is identified by an account qualifier.

Transfer Transactions

BASE24-atm supports the following transfer transactions.

Note: Subtype APT0 is used to identify a preferred transaction. .

Tran Code	Transaction Type and Description
40 01 01	**Transfer from Checking to Checking —** A transfer of funds from one of a cardholder's checking accounts to another.
40 01 11	**Transfer from Checking to Savings —** A transfer of funds from a cardholder's checking account to the cardholder's savings account.
40 01 60	**Transfer from Checking to Other —** A transfer of funds from a cardholder's checking account to another account identified by an account qualifier.
40 11 01	**Transfer from Savings to Checking —** A transfer of funds from a cardholder's savings account to the cardholder's checking account.
40 11 11	**Transfer from Savings to Savings —** A transfer of funds from one of a cardholder's savings accounts to another.
40 11 60	**Transfer from Savings to Other —** A transfer of funds from a cardholder's savings account to another account identified by an account qualifier.

Tran Code	Transaction Type and Description
40 31 01	**Transfer from Credit Account to Checking —** A transfer of funds from a cardholder's credit account to the cardholder's checking account.
40 31 11	**Transfer from Credit Account to Savings —** A transfer of funds from a cardholder's credit account to the cardholder's savings account.
40 31 60	**Transfer from Credit Account to Other —** A transfer of funds from a cardholder's credit account to the cardholder's account identified by an account qualifier.
40 60 01	**Transfer from Other to Checking —** A transfer of funds from a cardholder's account identified by an account qualifier to the cardholder's checking account.
40 60 11	**Transfer from Other to Savings —** A transfer of funds from a cardholder's account identified by an account qualifier to the cardholder's savings account.
40 60 60	**Transfer from Other to Other —** A transfer of funds from a cardholder's account identified by an account qualifier to another account identified by an account qualifier.

Payment Transactions

BASE24-atm supports the following payment transactions.

Tran Code	Transaction Type and Description
50 01 31	**Payment from Checking to Credit Account —** A transfer of funds, as a payment, from a cardholder's checking account to the cardholder's credit account.
50 11 31	**Payment from Savings to Credit Account —** A transfer of funds, as a payment, from a cardholder's savings account to the cardholder's credit account.

Tran Code	Transaction Type and Description
50 31 31	**Payment from Credit Account to Credit Account —** A transfer of funds, as a payment, from a cardholder's credit account to another credit account belonging to the same cardholder.
50 60 31	**Payment from Other to Credit Account —** A transfer of funds, as a payment, from a cardholder's account identified by an account qualifier to the cardholder's credit account.
51 00 00	**Payment Enclosed —** A payment toward outstanding debts, such as loans, made by deposit.

Check-Related Transactions

BASE24-atm supports the following check-related transactions.

Note: BASE24-atm self-service banking (SSB) allows a check to be deposited either in an envelope or directly into the ATM. The transaction code is the same for either method.

Tran Code	Transaction Type and Description
03 01 00	**Check Guarantee —** A cardholder can verify that the amount of the check is available in the checking account and place the amount of the check on hold. **Note:** Only the BASE24-atm Authorization process supports this transaction. This transaction is not supported by standard Device Handler, Interchange Interface, or Host Interface processes.
04 01 00	**Check Verification —** A cardholder can verify that the amount of the check is available in the checking account. **Note:** Only the BASE24-atm Authorization process supports this transaction. This transaction is not supported by standard Device Handler, Interchange Interface, or Host Interface processes.
11 00 00	**Cash Check —** A cardholder can deposit checks and receive that cash back immediately using this transaction.

Log-Only Transactions

BASE24-atm supports the following log-only transactions.

Tran Code	Transaction Type and Description
60 00 00	**Message to Financial Institution —** Written communication from the cardholder to the financial institution. Subtype AER0 is used for exchange rate notification processing. Subtype APS0 is used for preferred transaction setup processing.
61 01 00 61 02 00 61 03 00 61 04 00	**Log-Only Transaction —** Messages displayed as menu selections on the ATM screen. Cardholders can select specific messages to the institution with key selections at the ATM. Possible log-only transactions include "REORDER CHECKS" and "SEND MORTGAGE LOAN INFO."
62 00 00	**Card Review —** A verification of accounts, status, and card limits associated with a card. Only the Olivetti Device Handler and the BASE24-atm Authorization processes support this transaction; therefore, this transaction is available only with Olivetti terminals. Subtype API0 is used for preferred transaction inquiry processing.

Statement Print and Passbook Update Transactions

BASE24-atm supports the following statement print and passbook update transactions.

Note: Subtype APU0 is used for passbook update processing.

Tran Code	Transaction Type and Description
70 01 00	**Statement Print Transaction for Checking —** A request from a cardholder for a list of statement items posted to the cardholder's checking account.

Tran Code	Transaction Type and Description
70 11 00	**Statement Print Transaction for Savings Account —** A request from a cardholder for a list of statement items posted to the cardholder’s savings account.
70 31 00	**Statement Print Transaction for Credit Account —** A request from a cardholder for a list of statement items posted to the cardholder’s credit account.
70 60 00	**Statement Print Transaction for Other —** A request from a cardholder for a list of statement items posted to an account identified by an account qualifier.

PIN Change Transaction

BASE24-atm supports the following PIN change transaction.

Tran Code	Transaction Type and Description
81 00 00	**PIN Change Request —** A request from a cardholder to change the Personal Identification Number (PIN) associated with his or her card.

EMV PIN Unblock Transaction

BASE24-atm supports the following PIN unblock transaction:

Tran Code	Transaction Type and Description
82 00 00	**EMV PIN Unblock Request —** A request from a cardholder to remove the block from the Personal Identification Number (PIN) associated with his or her card.

Administrative Transactions

BASE24-atm supports the following administrative transactions entered from ATMs directly connected to the system. Administrative transactions that are sent to the BASE24 Authorization process must be performed using a card number that

BASE24 has been configured to recognize as belonging to an administrative card. This may be a physical card that is used at the front of the terminal or a number that is specified in the ATM's configuration data for rear administrative requests.

When the Authorization process determines that a transaction is an administrative request, it examines the first two digits of the processing code for a value of 98, which indicates that the request was initiated at the rear of the ATM.

If the ATM-REAR-ADMN-TRAN-PIN-ALWD param is present in the LCONF and has the value N, the Authorization process will deny any rear administrative transaction that contains a PIN. Payment Card Industry (PCI) standards strongly discourage entering PINs at the rear of an ATM because of the security risk. If the ATM-REAR-ADMN-TRAN-PIN-ALWD param is not present in the LCONF or does not have the value N, or if the transaction does not contain a PIN, the Authorization process will handle the rear administrative request identically to a front administrative request. Except for the special transaction code of 98, the processing code in an administrative request is specific to the Device Handler and is not used by the Authorization process.

When the Authorization process receives an administrative request that was initiated with an administrative card, the Authorization process identifies the card type as administrative, verifies whether the transaction is an on-us transaction, and determines the appropriate response code depending on the balance flag and balancing window.

Only certain device handlers, such as the Diebold 10XX/478X and NCR 5XXX Device Handlers, send administrative requests to the Authorization process for approval. Administrative requests from the Device Control Terminal (DCT) and from other Device Handlers are not sent to the Authorization process for approval.

Tran Code	Transaction Type and Description
-- -- --	**Terminal Balancing —** A request from an ATM service person to clear the ATM totals and close out the processing day.
-- -- --	**Mid-Point Cash Adjustment —** A request from an ATM service person to change the ATM hopper totals to reflect an addition or removal of cash from a hopper.
-- -- --	**Terminal Sub-Total —** A request from an ATM service person for a listing of the ATM totals.

BASE24-atm Message Types

All transactions are carried out by means of messages exchanged between the originator and authorizer of the transaction. However, mobile top-up transactions can be routed to multiple secondary routing destinations based on the transaction subtype used in the Split Transaction Routing File (STRF). These messages are assigned numbers to identify their message type and function.

Message Functions

Messages can be grouped into the following functional categories.

Request — A request from a transaction originator to a transaction authorizer to authorize a financial transaction.

Response — A response from a transaction authorizer to a transaction originator approving or declining a financial transaction request.

Advice/Force Post — A message from a transaction originator to a transaction authorizer so the authorizer can update its files. Advices are also known as force post transactions because the authorizer must accept the transactions.

Reversal — A message from the transaction originator to the transaction authorizer indicating that any database impacts from an earlier transaction should be reversed because the transaction either could not be completed or was completed incorrectly.

Adjustment — A message from a transaction originator to a transaction authorizer requesting that the amount of a previously approved transaction be adjusted or corrected.

Notification/Selection — A message from a transaction authorizer to a transaction originator when actions are required by the customer to either acknowledge a surcharge/tax or to make a selection in order for the transaction to continue.

Message Types

Message types are used inside and outside of the BASE24-atm system. The following table identifies the message types used inside BASE24-atm along with the message type as they would be sent to, or received from, a host or a BIC co-network. The internal message types are defined in the table column identified by INT. External message types are provided for both ISO and ANSI communications with hosts and BIC co-networks. These external message types are identified in the table in the ISO and ANSI columns.

INT	ISO	ANSI	Description
0200	0200	0200	**Financial Transaction Request —** A request to authorize a transaction.
0205	0205	0205	**Statement Print Transaction Request —** A request to print statement information. Note that the BIC ISO Interface does not support this transaction. Passbook update transactions enter BASE24-atm as statement print transactions with the subtype APU0.
0210	0210	0210	**Financial Transaction Request Response —** An approval or denial response to an authorization request.
0215	0215	0215	**Statement Print Transaction Response —** A response carrying the statement information for printing. Note that the BIC ISO Interface does not support this transaction. Passbook update transactions enter BASE24-atm as statement print transactions with the subtype APU0.
0220	0220	0220	**Financial Transaction Advice —** A completion message generated as notification to a host or co-network that BASE24-atm has approved a transaction. When BASE24-atm receives a 0220 message, it considers it a “force post” transaction.
Not used	0221	0221	**Financial Transaction Advice Repeat —** A reissue of the completion message (0220) generated to notify a host or co-network that BASE24-atm has approved a transaction.

INT	ISO	ANSI	Description
Not used	0230	0230	**Financial Transaction Advice Response —** An acknowledgment of receipt generated by a host or co-network in response to a financial transaction advice (0220). This message type can also be sent by an Interchange Interface process.
Not used	0231	0231	**Financial Transaction Advice Response Repeat —** A reissue of the acknowledgment of receipt generated by a host or co-network in response to a financial transaction advice (0220).
0420	0420	0400	**Reversal —** A message indicating that a transaction was not completed as authorized.
Not used	0421	0401	**Reversal Repeat —** A reissue of the reversal message (0420) indicating that a transaction was not completed as authorized.
Not used	0430	0410	**Reversal Response —** An acknowledgment of receipt generated by a host or co-network in response to a reversal (0420). This type can also be sent by an Interchange Interface process.
Not used	0800	0800	**Network Management Message Request —** A request to perform echo-test, logon, and logoff.
Not used	0810	0810	**Network Management Message Response —** An acknowledgment of receipt generated in response to a network management message request (0800).
5400	5400	5400	**Financial Transaction Correction —** An adjustment to a previously completed transaction. An adjustment (5400) can be sent by an Interchange Interface process.
Not used	Not used	9220	**Store-and-Forward Financial Transaction Advice —** A financial transaction advice or reversal that is being sent to the host from the BASE24 Store-and-Forward File (SAF). When an advice cannot be sent to a host because communications to the host are down, the advice is stored in the SAF for sending when host communications are restored.

INT	ISO	ANSI	Description
Not used	Not used	9221	**Store-and-Forward Financial Transaction Advice Repeat —** A reissue of a store-and-forward financial transaction advice or reversal (9220). A store-and-forward financial transaction advice repeat is identical to the original advice it duplicates, except that it denotes to the receiver that it is a possible duplicate message. A store-and-forward financial transaction advice repeat is used when an acknowledgment was not received for a store-and-forward advice.
Not used	Not used	9230	**Store-and-Forward Financial Transaction Advice Response —** An acknowledgment of receipt generated by a host in response to a store-and-forward financial transaction advice (9220).
Not used	Not used	9231	**Store-and-Forward Financial Transaction Advice Response Repeat —** A reissue of an acknowledgment of receipt (9230) generated by a host in response to a store-and-forward financial transaction advice.
9901	Not used	Not used	**Surcharge Notification —** A message indicating an acquirer fee (surcharge) is involved in the transaction. Acknowledgment of the surcharge is required by the cardholder.
9906	Not used	Not used	**Account Data Selection —** A message containing account data for selection by a cardholder. BASE24 uses 9906 messages when selections are required for transactions involving multiple accounts.
9907	Not used	Not used	**Card Review Account Selection —** A response message containing all accounts associated with a card. This message is used by the Olivetti Device Handler process only.
9909	Not used	Not used	**Mobile Top-Up Notification/Selection —** A message containing mobile top-up data for acknowledgment/selection by a cardholder. BASE24 uses 9909 messages when an acknowledgment/selection is required for transactions involving mobile top-up activity.

INT	ISO	ANSI	Description
9980	Not used	Not used	**Informational (Money in Drawer) —** An informational message advising the BASE24-atm system that the cardholder left money in the drawer. This message is used by the Diebold, Fujitsu, and NCR Device Handler processes only.
9991	Not used	Not used	**Log-Only Transaction —** A message that must be logged to the Transaction Log File (TLF).

Transaction Logging

The BASE24-atm Authorization process logs all of the transactions it processes to the Transaction Log File (TLF). The Interchange Interface processes and BIC Interface processes log all the transactions they process to the appropriate Interchange Log Files (ILFs) if configured to do so.

Message Types Logged

The Authorization and the Interchange Interface processes convert messages from the internal message into the TLF or the ILF format before logging them to the TLF or the ILF. The following BASE24-atm message types are logged to the TLF or the ILF:

Type	Description
0210	Authorization response
0215	Statement print response—first only Passbook update transactions with the subtype APU0 are logged to the TLF.
0220	Advice—incoming only
0420	Reversal (refer to note below)
5400	Transaction adjustment (refer to note below)
9980	Informational (money in drawer)
9991	Log-only transaction **Note**: If the TLF file is unavailable, log the ATM settlement and balancing messages to the EMS log.

Note: Neither the 0420 or 5400 message is logged to the ILF. Both messages are used to update existing 0210 and 0220 records that have already been written to the ILF.

The following BASE24-atm message types are not logged to the TLF or ILF:

Type	Description
0200	Authorization request
0205	Statement print transaction request
0220	Advice—created locally (that is, completion)
9901	Surcharge notification
9906	Account data selection
9907	Card review account selection
9909	Mobile top-up notification/selection

Tokens Logged

The Token File (TKN) controls which tokens in the internal message are logged to the TLF or ILF. All tokens that can be logged to the TLF or ILF are displayed on TKN screen 2. As each message is logged to the TLF or ILF, tokens identified to be logged are arranged in the TLF or ILF record in the order in which they were added to the internal message.

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2: BASE24-atm Transaction Path

A number of processes and processing entities form the BASE24-atm transaction path—that is, the series of BASE24-atm processes and processing entities through which a specific transaction passes in order to be authorized.

The actual path each transaction takes varies based on a number of factors, including the type of transaction being processed, the transaction originator, the transaction authorizer, and the processing set up by the BASE24 processor.

This section discusses the following aspects of the BASE24-atm transaction path:

- BASE24-atm processes that can be part of the transaction path.
- Process linkages (a description of how the various BASE24-atm processes and processing entities are linked together for the purpose of forwarding and receiving transaction messages).
- Service-based routing (a description of service-based routing within BASE24-atm).
- Timers (a description of the various timers used by BASE24-atm processes to ensure the timely completion of transactions).

BASE24-atm Processes

Along the transaction path, processing is carried out in stages by several BASE24-atm processes, each of which has its own responsibilities. The BASE24-atm processes that can be involved in transaction processing are listed below and described on the following pages:

- Device Handler processes
- Authorization processes
- Host Interface processes
- Interchange Interface processes
- BIC Interface processes
- Transaction Context Manager process

BASE24-atm processes all communicate with one another through the XPNET process.

Device Handler Processes

BASE24-atm Device Handler processes are responsible for communicating with ATMs (that is, devices) directly attached to BASE24-atm. Device Handler processes are designed specifically for each type of device; however, all Device Handler processes perform essentially the same functions which include the following:

- Recognizing native-mode messages sent from the ATM and translating them into the BASE24 internal message format.

  Each type of ATM sends and receives messages in a device-specific format called native mode. BASE24-atm is not capable of using the various native-mode message formats internally. Therefore, the Device Handler process translates the messages between the ATM and BASE24-atm.

- Determining whether the transaction needs to be authorized by the BASE24-atm Authorization process or sent to a different authorization destination. For example, a bill payment transaction might be authorized by a bill payment service running under the same XPNET system.

  Note: The APCF routing functionality is supported for the Diebold 10XX/478X and NCR 5XXX Device Handler processes only.

In addition, mobile top-up transactions are sent to the Transaction Context Manager process for split routing, unless enhanced mobile top-up routing support is being used.

- Setting and stopping transaction response timers as transactions pass in and out of the Device Handler process.

  These response timers control the maximum length of time the Device Handler process waits for a response from the Authorization process before cancelling the transaction. For example, when the Device Handler process receives a transaction from a terminal, it sets a response timer and then passes the transaction on for processing. When processing completes the Device Handler process deletes the timer. If the Device Handler does not receive a response before the timer expires, the transaction is declined.

  Also, if the transaction is declined because the Device Handler timer expired, and if the Device Handler process receives a late response from the Authorization process, the Device Handler process generates an appropriate reversal.

- Updating the accumulator fields and status flags in the BASE24-atm Terminal Data files record for each ATM.

  When transactions pass through the Device Handler process, the Device Handler process updates the BASE24-atm Terminal Data files accumulation fields and other BASE24-atm Terminal Data files processing fields and flags depending on the type of transaction and processing required.

- Cutting over terminals for daily settlement using either a terminal balancing transaction or a Device Handler force cutover option.

  If the Device Handler process does not cut over the terminal, the Settlement Initiator process cuts the terminal over. Refer to the ***BASE24-atm Settlement and Reporting Manual*** for more information about terminal cutover.

- Facilitating PIN and message authentication support between the terminal and BASE24-atm. For information on the PIN and message authentication support provided by BASE24-atm Device Handler processes, refer to the ***BASE24 Transaction Security Manual***.

- Downloading terminal configuration information to ATMs.

  The information is downloaded from terminal-specific configuration and load files.

Authorization Processes

BASE24-atm Authorization processes are responsible for many transaction processing tasks. These tasks include the following:

- Authorizing or routing transactions to the appropriate interface process, which sends the transaction to the entity that is to authorize the transaction.
- Logging transactions to the Transaction Log File (TLF).
- Generating completion messages to notify hosts and co-networks regarding transactions approved by BASE24-atm.
- Calculating surcharge fees when applicable.

Host Interface Processes

BASE24-atm Host Interface processes are responsible for communicating with host systems. They become involved in processing when the host is the appropriate authorizer or is to receive completion messages for transactions authorized by BASE24-atm. They also become involved in processing when BASE24 is the authorizer, and the host is the transaction acquirer.

There are two types of Host Interface processes: one that communicates with hosts using an ANSI-based message format and one that communicates with hosts using an ISO-based message format. For information on the ANSI Host Interface process, refer to the ***BASE24 ANSI Host Interface Manual.*** For information on the ISO Host Interface process, refer to the ***BASE24 ISO Host Interface Manual*** and ***BASE24 External Message Manual.***

The Host Interface process performs the following functions:

- Translating messages from the BASE24-atm internal message format to the ISO or ANSI external format recognizable to a host.
- Providing a store-and-forward function, whereby if a host is unavailable, certain types of messages bound for the host can be stored and sent to the authorizing entity at a later time.
- Setting and stopping transaction timers that monitor the time it takes for messages to return to the Host Interface process from another entity.
- Providing for PIN, MAC, and key management between BASE24-atm and the host. For information on the transaction security functions performed by Host Interface processes, refer to the ***BASE24 Transaction Security Manual***.

Interchange Interface Processes

BASE24-atm Interchange Interface processes are responsible for communicating with interchanges. They become involved in processing when the interchange is the appropriate authorizer or when BASE24 is the authorizer, and the interchange is the transaction acquirer.

Each interchange has its own Interchange Interface process because each interchange has its own message format and message handling requirements. Interchange Interface processes are typically responsible for the following:

- Translating messages between the BASE24-atm internal message format and the external message format recognizable to the interchange.
- Determining whether outbound transactions are allowed to be sent from BASE24 to the interchange.
- Determining whether inbound transactions are allowed from the interchange, and if so, whether they need to be authorized by BASE24-atm or sent to a different authorization destination. For example a bill payment transaction might need to be authorized by a bill payment service.

  Note: Inbound transaction allowed checking and APCF routing functionality is supported for the following Interchange Interface processes only:

  - Banknet Interchange Interface process
  - BIC ISO Interface process
  - MDS/MDSM Interchange Interface process
  - PLUS ISO Interchange Interface process
  - VisaNet Interchange interface process
- Logging transactions to the Interchange Log File (ILF).
- Providing a store-and-forward function, whereby, if the interchange is unavailable, certain types of messages bound for the interchange can be stored and sent to the interchange at a later time.
- Setting and stopping transaction timers that monitor the time it takes for messages to return to the Interchange Interface process from another entity.
- Providing for PIN, MAC, and key management between BASE24-atm and the interchange. For information on the transaction security functions performed by Interchange Interface processes, refer to the ***BASE24 Transaction Security Manual***.

Transactions entering BASE24-atm from an interchange must first pass through the Interchange Interface process. The Interchange Interface process translates the message from the interchange external message to the ATM Standard Internal Message (STM) before routing the message to the other BASE24 processes.

Transactions bound for an interchange that originated at a host or at an ATM connected to BASE24-atm must first pass through the appropriate Interchange Interface process. The Interchange Interface process translates the message into the interchange external message format before sending it to the interchange.

BIC Interface Process

BASE24 Interchange (BIC) Interface processes are responsible for communicating with other BASE24 systems and co-networks. They become involved in processing when a transaction must be forwarded to another BASE24 system or co-network for authorization or a transaction is being sent from another BASE24 system or co-network for authorization.

There are two types of BIC Interface processes: one that communicates with co-networks using an ANSI-based message format and one that communicates with co-networks using an ISO-based message format. For information on the BIC ANSI Interface process, refer to the ***BASE24 BIC ANSI Interface Manual.*** For information on the BIC ISO Interface process, refer to the ***BASE24 BIC ISO Interface Manual*** and ***BASE24 BIC ISO Standards Manual.***

The BIC Interface process is responsible for the following:

- Translating messages between the BASE24-atm internal message format and the external message format recognizable to the BIC co-network.
- Determining whether outbound transactions are allowed to be sent from BASE24 to the interchange.
- Determining whether inbound transactions are allowed from the interchange, and if so, whether they need to be authorized by BASE24-atm or sent to a different authorization destination. For example a bill payment transaction might need to be authorized by a bill payment service.

  Note: Inbound transaction allowed checking and APCF routing functionality is supported for the BIC ISO Interface process only.
- Logging transactions to the Interchange Log File (ILF).

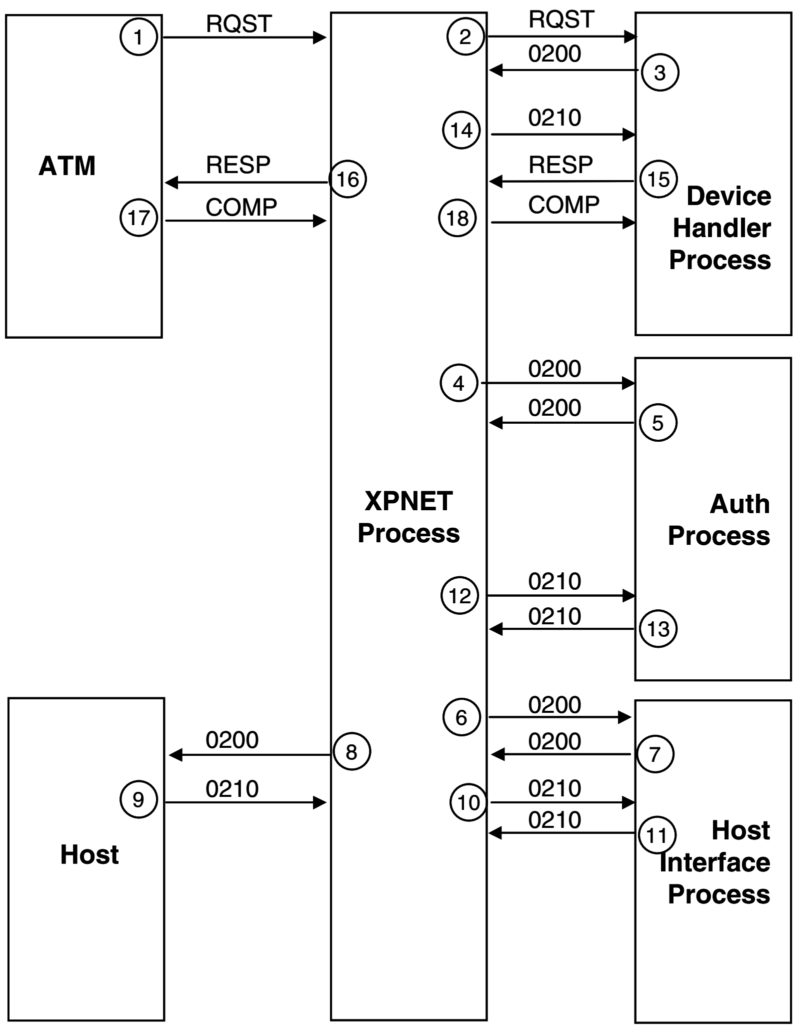
Steps	Processing
1, 2	The interchange sends a request message to the Interchange Interface process.
3, 4	The Interchange Interface process reformats the interchange message into an internal 0200 request message, sets an inbound request timer, and sends the 0200 request message to the Authorization process.
5, 6	The Authorization process performs the following steps: a. Uses the Card Prefix File (CPF) to locate an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 2. The Authorization process approves the request, logs the response to the Transaction Log File (TLF), updates the database files appropriately, and returns a 0210 response message to the Interchange Interface process. Processing continues with step 7a. b. Since the host requires completions, the Authorization process sends a 0220 completion message to the Host Interface process when the COMPLETION REQUIRED TO HOST field on the IPCF screen is set to a nonzero value. Processing continues with step 7b.
7a, 8a	The Interchange Interface process reformats the 0210 response message into a response message understood by the interchange, logs the response in the Interchange Log File (ILF), deletes its inbound request timer, and sends the response message to the interchange.
7b, 8b	The Host Interface process reformats the 0220 internal message into an external message and sends it to the host.

BASE24-atm Transactions: Authorization Level 3

The following pages contain message flows for transactions being processed by BASE24-atm using level 3 (online/offline) authorization processing. This authorization level requires that transactions be sent to a host for authorization, but allows BASE24-atm to stand in and authorize transactions when the host is not available.

Approved by Host — Completions Not Required

The diagram below illustrates the flow of a transaction that originates at an ATM and is approved by a host. The host does not require completions.



Steps	Processing
1, 2	The ATM sends a native-mode request message to the Device Handler process.
3, 4	The Device Handler process reformats the native-mode message into a 0200 request message, sets an authorization timer, and sends the 0200 request message to the Authorization process.

Steps	Processing
5, 6	From the Card Prefix File (CPF), the Authorization process locates an Institution Definition File (IDF) record for the institution that issued the card. The IDF record identifies how the transaction request is to be processed. In this case, the transaction request is to be processed using authorization level 3. The Authorization process sends the 0200 request message to the Host Interface process.
7, 8	The Host Interface process reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to the host.
9, 10	The host approves the request and returns a 0210 response message to the Host Interface process.
11, 12	The Host Interface process reformats the external message into an internal message, deletes its outbound request timer, and sends the 0210 response message to the Authorization process.
13, 14	The Authorization process logs the response to the Transaction Log File (TLF), updates the database files, and sends the 0210 response message to the Device Handler process.
15, 16	The Device Handler process deletes its authorization timer and reformats the 0210 response message into a native-mode response that it sends to the ATM.
17, 18	The ATM sends a native-mode message to the Device Handler process to indicate that the transaction completed as authorized.

Approved by BASE24-atm — Completions Required — Completion Placed in Store-and-Forward File

The diagram below illustrates the flow of a transaction that originates at an ATM and is approved by BASE24-atm, which stands in for the host because communications to the host are down. The host requires completions; however, since communications are down, BASE24-atm places the completion message in the Store-and-Forward File (SAF) to be sent to the host at a later time.